LANCERSNAP

Software Design Document

AI-Assisted CBU Campus Reporting Application

CSCI 5520 - AI-Driven Software Development

Document Information	Details
Document Version	v0.2
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Institution	California Baptist University
Project Repository	https://github.com/JudahAzariah/LancerSnap
Primary Product Scope	Photo-first campus issue reporting, AI-assisted classification and routing, staff review, and report resolution

Design principle

A student should be able to report a campus issue with a photo, a location, and minimal additional effort. AI assists with organization; authorized staff make operational decisions.

Document Guide

This document defines the LancerSnap problem, users, MVP scope, requirements, architecture, user experience, and durable design decisions. Requirement identifiers are stable references for the Implementation Plan and GitHub Issues.

Section	Contents
1. Problem and Users	Problem statement, goals, users, and stakeholders.
2. Proposed Solution and Scope	Product description, MVP capabilities, and explicit non-goals.
3. User Stories and Acceptance Criteria	US-01 through US-13.
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5. Platform and Architecture	Technology choices, system components, and data flow.
6. UI/UX Design	Student, staff, and administrator experiences.
7. Design Decisions, Assumptions, and Risks	DD-01 through DD-10, open decisions, and risks.
8. References	Supporting platform documentation and project artifacts.

1. Problem and Users

1.1 Problem Description

Students at California Baptist University (CBU) encounter physical and operational campus issues, including facility problems, potential safety hazards, housing-related concerns, technology problems, and issues affecting campus services. A student may not know which campus team is responsible, what information staff need, or where to send a report. This uncertainty can delay reporting and create avoidable back-and-forth for both students and staff.

A reporting process that relies on students selecting the correct office and manually composing a detailed description places too much work on the person reporting the issue. Staff may also receive incomplete submissions, duplicate reports about the same issue, or reports that are not organized by location, category, and urgency.

After submitting a report, students need a clear indication that it was received and a way to understand whether staff have assigned, addressed, or resolved the issue. Staff need a manageable queue with useful context and control over the decisions that affect real campus operations.

> **Core Problem** CBU needs a centralized, low-friction campus reporting process that lets students report an issue primarily by taking a photo, while helping authorized staff classify, route, prioritize, track, and resolve submissions.

1.2 Product Goals

- Make submitting a campus issue quick and straightforward, with a photograph as the primary input and only minimal required information.
- Use AI to recommend report categories, extract useful details, suggest priority, and identify potential duplicates.
- Route reports using configurable rules that reflect CBU operational teams rather than requiring students to know the organization chart.
- Give authorized staff a clear way to review reports, make decisions, update status, and resolve issues.
- Give students transparent confirmation and progress information after submission.
- Protect report data and uploaded media through authentication, authorization, and controlled storage.
- Provide basic aggregated reporting information to authorized administrators.

1.3 Target Users and Stakeholders

User / Stakeholder	Role in LancerSnap
CBU Student	Primary reporter. Submits a photo-based campus issue report, provides location context, and tracks progress.
Campus Staff	Reviews reports routed to an authorized team queue, evaluates AI suggestions, updates status, and resolves issues.
LancerSnap Administrator	Manages supported categories, routing rules, access configuration, and report-trend views.
CBU Operational Teams	Existing campus teams or roles that receive reports according to validated routing configuration. Examples considered include Housing/Residential Assistants, Facilities, Dining, Building Administration, and IT/Technology Support.
University Leadership	May review aggregated operational trends to understand recurring campus issues.

The examples of operational teams are routing candidates, not a declaration of official CBU ownership. The actual category-to-team mapping must be confirmed with CBU before production use.

2. Proposed Solution and Scope

2.1 Proposed Solution

LancerSnap is an AI-assisted campus reporting application for California Baptist University. It is designed around a simple student action: take or upload a photograph of a campus issue, select or confirm where it occurred, optionally add a short note, and submit the report.

The student is not required to choose the responsible campus office. After a report is submitted, the backend can request AI analysis to suggest a category, extract useful visible details, recommend a priority, and identify potentially similar reports. Configurable routing rules then direct the report to an appropriate team queue. Authorized staff review the submitted information and decide what action to take.

Students can track their submissions and receive configured status updates. Staff can manage assigned reports through a dashboard. Administrators can maintain supported categories and routing configuration and view aggregated report trends.

Primary workflow Student takes photo -> confirms campus location -> optionally adds a note -> submits -> AI recommends category/details/priority -> configured routing selects a team queue -> staff reviews and acts -> student sees status and resolution updates.

2.2 Minimum Viable Product (MVP) Scope

Student capabilities

- Authenticate using an authorized account.
- Open the primary Snap & Report experience.
- Take a photograph or select an existing photograph for a report.
- Select or confirm a supported CBU campus building or location.
- Add an optional short description or note.
- Submit a report without selecting the responsible campus office.
- Receive a clear submission acknowledgement and report identifier.
- View a list of their own reports and open report details.
- Track report status and receive configured resolution/status updates.
- Reopen a report when the issue remains unresolved, subject to the defined workflow.

AI-assisted capabilities

- Recommend a category from the configured set of campus issue categories.
- Generate a concise report summary and extract useful details from the submitted image and optional note.
- Recommend a priority level, with potential safety hazards clearly flagged for human attention.
- Identify potential duplicate reports using available report attributes such as category, location, submission time, text, and image similarity where supported.
- Provide structured output and uncertainty information so downstream services and staff can distinguish suggestions from confirmed facts.
- Support routing through administrator-configured rules; AI output does not independently authorize consequential action.

Campus staff capabilities

- Access reports assigned to their authorized team or office queue.
- Review submitted photographs, optional notes, location, report history, and AI-generated information.
- Review suggested priority and potential duplicate information.
- Accept or assign reports within the configured staff workflow.
- Update permitted report statuses and record resolution information.
- Reassign a report when the configured destination is not appropriate, subject to authorization.

Administrator capabilities

- Create, update, or deactivate supported report categories.
- Configure category/location-based routing rules for operational teams.
- Manage authorized users and role assignments within the approved access model.
- View aggregated report trends by supported category, location, status, or time period.

2.3 Supported Report Categories

The MVP uses configurable categories for physical and operational campus issues. The initial category set should be approved and maintained by an authorized administrator. Candidate categories aligned with the implementation plan include:

- Facilities / maintenance issue
- Potential safety hazard
- Housing / residential issue
- IT / technology problem
- Dining or campus service-location issue
- Other supported campus issue

The category list and its mapping to operational teams are configuration data. Do not hard-code unverified CBU department names, personal contacts, or assumed procedures into the application.

2.4 Out of Scope and Non-Goals

The following capabilities are not part of the current MVP:

- Harassment or disciplinary reporting workflows.
- Sensitive personal complaint workflows.
- Direct integration with 911, police dispatch, fire/EMS, or other emergency services.
- Replacing CBU work-order, facilities-management, or case-management systems.
- Facial recognition or identification of individuals from submitted photographs.
- Continuous GPS tracking of students.
- Automated administrative, disciplinary, safety, or other consequential decisions.
- Full campus-wide integration with every university information system.
- Advanced predictive analytics or long-term campus forecasting.
- Native desktop applications.

Scope boundary LancerSnap is a campus issue reporting and workflow-support tool. It does not replace emergency

services, existing university procedures, or the judgment of authorized campus personnel.

3. User Stories and Acceptance Criteria

The following stories define the observable behaviors required by the MVP. The identifiers are stable references for the Implementation Plan, GitHub Issues, and project-context documentation.

3.1 Student

US-01 - Submit Report

User Story: As a student, I can submit a photograph with an optional note and a campus location so that I can quickly report an issue.

Acceptance Check: Pass if a valid submission is stored, receives a unique report identifier, and displays a receipt/confirmation.

US-02 - AI Classify Report

User Story: As a student, I can see the suggested category for my report so that I understand how the issue was interpreted.

Acceptance Check: Pass if the analysis returns a supported category in a structured response and the interface labels it as an AI suggestion.

US-03 - Track Report

User Story: As a student, I can view the current status of my submitted report so that I understand its progress.

Acceptance Check: Pass if the student can view only their own reports and see the latest authorized status.

US-04 - Receive Resolution Update

User Story: As a student, I receive an update when staff resolve my report so that I know the issue has been addressed.

Acceptance Check: Pass if a resolved status is saved and the configured student-facing update is generated; notification delivery failure is visible and does not falsify the saved status.

US-05 - Reopen Report

User Story: As a student, I can reopen a report when the issue remains unresolved so that staff can review it again.

Acceptance Check: Pass if the permitted reopen action changes the report state, records the event, and returns it to the appropriate authorized queue.

US-06 - Report Location

User Story: As a student, I can select or confirm the campus location where the issue occurred so that staff receive useful location context.

Acceptance Check: Pass if a supported campus location is associated with the report and is available to authorized staff.

3.2 Campus Staff

US-07 - View Assigned Reports

User Story: As authorized campus staff, I can view reports routed to my team so that I can process them.

Acceptance Check: Pass if staff can access only reports within their authorized team/office scope.

US-08 - Review AI Information

User Story: As authorized campus staff, I can review AI-generated classification, extracted details, priority suggestions, and potential duplicate information so that I can make an informed decision.

Acceptance Check: Pass if available AI information is visible to authorized staff and is clearly distinguished from confirmed staff decisions.

US-09 - Update Report Status

User Story: As authorized campus staff, I can update a report status so that the student can track its progress.

Acceptance Check: Pass if an allowed status transition is saved by the backend, attributed to an authorized actor, and shown in the student report view.

US-10 - Resolve Report

User Story: As authorized campus staff, I can mark an issue as resolved and record resolution information so that the reporting workflow can be completed.

Acceptance Check: Pass if an authorized staff member can resolve a report, the resolution is recorded, and the student can see the resulting state.

3.3 Administrator

US-11 - Manage Routing

User Story: As an administrator, I can configure category- and location-based routing rules so that reports reach an appropriate operational team.

Acceptance Check: Pass if an authorized administrator can save a routing rule and the routing engine applies the active rule to matching reports.

US-12 - View Report Trends

User Story: As an administrator, I can view aggregated report trends so that recurring campus issues can be identified.

Acceptance Check: Pass if reports can be grouped by at least one supported category, location, status, or time period.

US-13 - Manage Categories

User Story: As an administrator, I can create, update, or deactivate supported report categories so that classification and routing use the approved category set.

Acceptance Check: Pass if an authorized administrator can modify category configuration and the application uses the active configuration without changing the student-facing workflow.

4. Requirements, Constraints, and Design Rules

These requirements apply across the application and should be treated as implementation constraints. They are referenced by identifier in the Implementation Plan and GitHub Issues.

RC-01 - Access Control: Students may access only their own reports. Staff may access reports assigned to their authorized team or office scope. Administrative functions require administrator authorization. Backend authorization is authoritative; hiding interface controls is not sufficient.

RC-02 - AI Uncertainty: AI-generated categories, extracted details, summaries, priority suggestions, and duplicate indications must be treated as recommendations. The system must preserve or display uncertainty when the model cannot make a reliable determination.

RC-03 - Human Oversight: Authorized human staff retain final responsibility for operational actions, report assignment, status changes, resolution, and decisions involving safety. AI output must not independently complete consequential actions.

RC-04 - Configurable Routing: Routing rules must be configurable by an authorized administrator. Category-to-team behavior must not be permanently embedded in the student interface or scattered as hard-coded conditions throughout the application.

RC-05 - Team-Based Routing: Reports must be routed to configured organizational teams or role-based queues rather than depending permanently on an individual employee. The actual CBU team mapping must be validated before production deployment.

RC-06 - Duplicate Handling: The system may identify potentially similar reports using available attributes. A duplicate suggestion must not automatically merge, delete, or close reports. Staff must be able to review the suggestion and retain separate reports when appropriate.

RC-07 - Safety Priority: Potential safety hazards must receive a clearly distinguishable high-priority indication and be surfaced for appropriate human attention. AI priority is advisory and does not replace established emergency or campus safety procedures.

RC-08 - Emergency Boundary: LancerSnap must communicate that it is a campus reporting tool and does not provide emergency dispatch. The interface must direct users to contact the appropriate emergency service or campus emergency channel when immediate assistance is required.

RC-09 - Secure Media: Uploaded photographs must be stored in protected object storage and must not be publicly accessible by default. Access to original media must follow report authorization rules.

RC-10 - Error Handling: Failures in AI analysis, database operations, routing, storage, or notifications must be handled explicitly. The system must not display a false success state. A valid report submission must remain possible when an external AI service is unavailable; analysis can be retried or sent for human review.

RC-11 - Accessibility: Core reporting, tracking, and staff-management functions must use clear labels, readable text, sufficient contrast, and controls that do not rely on color alone to communicate important information. Core actions must be usable with accessible interaction patterns.

RC-12 - Status Transparency: Students must be able to see whether a report has been received and its current permitted status. Status changes must be persisted by the backend and communicated consistently in report details and configured notifications.

5. Platform and Architecture

5.1 Architectural Overview

LancerSnap uses a mobile-first client and cloud-hosted backend. The mobile application provides the student reporting experience and authorized staff/administrator interfaces. The backend coordinates authentication context, report creation, AI requests, routing, status transitions, and data access. Data is persisted in PostgreSQL, while uploaded photographs are stored separately in protected object storage.

The AI service is an external processing dependency. It returns structured recommendations to the backend; it is not the system of record and does not directly authorize staff actions. The routing engine applies administrator-configured rules to determine the destination team queue.

5.2 Technology Choices

Technology / Platform	Purpose in the System	Status / Boundary
React Native + Expo	Cross-platform mobile application for student, staff, and administrator experiences.	Selected client approach.
FastAPI (Python)	Backend APIs, application business logic, report management, routing orchestration, and integration with external services.	Selected backend approach.
PostgreSQL	Persistent relational storage for users, roles, reports, categories, locations, routing rules, status history, and related metadata.	Database engine.
Supabase	Managed authentication, PostgreSQL services, and protected object storage.	Selected managed platform for the prototype.
External multimodal AI service	Analyze submitted image and optional text to suggest category, summary/details, priority, and potential duplicates.	Provider/model remains an open decision.
Notification service	Deliver configured report-status updates to students.	Provider and exact delivery channel remain open.
Campus location reference data	Provide supported building/location choices during report submission.	Use selected campus locations; do not implement continuous GPS tracking.

5.3 Major System Components

Component	Responsibility
Mobile Application	Provides the photo-first student reporting flow, report tracking, and role-restricted staff and administrator interfaces.
Authentication and Authorization	Establishes user identity and enforces student, staff, and administrator permissions.

Component	Responsibility
Backend API	Validates requests, creates reports, coordinates services, enforces business rules, and exposes authorized report operations.
AI Processing Adapter	Sends approved report content to the selected AI provider, validates structured responses, and records AI output and uncertainty.
Duplicate Detection	Evaluates report attributes to identify potential overlap and returns suggestions for staff review.
Routing Engine	Applies active category/location/team rules and assigns a destination queue or invokes a manual-review fallback.
Report Workflow Service	Enforces allowed status transitions, assignment, resolution, reopening, and status history.
Database and Protected Media Storage	Persists application records and stores uploaded photographs with access restrictions.
Notification Service	Creates status-update notifications based on persisted report events.
Administrator Configuration and Analytics	Maintains categories/routing configuration and produces supported aggregate views.

5.4 System Architecture and Reporting Workflow

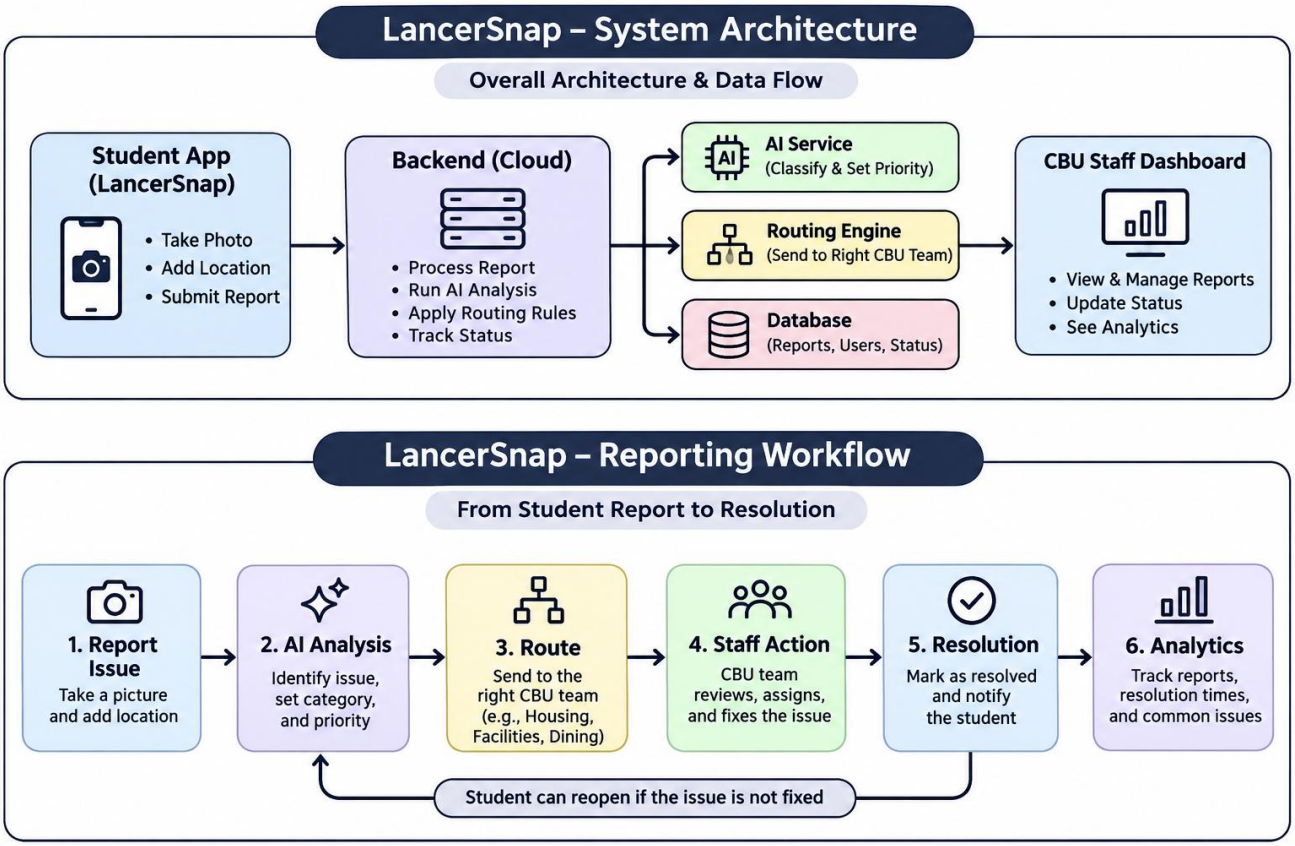


Figure 1. LancerSnap system architecture and reporting workflow.

5.5 Primary Data Flow

1. The student signs in and opens Snap & Report.
2. The student captures or selects a photograph, confirms a supported campus location, and optionally adds a note.
3. The mobile client submits the report to the backend. The backend validates the user, input, location, and media reference.

4. The backend stores the report and protected photograph reference and returns a report identifier/receipt.
5. The backend requests AI analysis when the AI service is available. The response is schema-validated and stored as advisory information.
6. The routing engine applies the active category/location/team configuration. If classification or routing is uncertain, the report enters the configured manual-review path.
7. Authorized staff review the report, AI suggestions, and any duplicate indication; staff decide what action to take.
8. Staff update report status or resolve the issue. The backend validates the transition and records status history.
9. The student sees the current status and receives configured status notifications. Notification failure does not erase the saved status.

5.6 Conceptual Data Areas

The logical data model should support the following areas. Exact table names, field types, and relationships are to be defined during database implementation and recorded in the repository architecture documentation.

Data Area	Representative Information
Users and roles	Authenticated user identifier, role, and authorized team/office membership.
Reports	Report identifier, submitter, category, description/note, location, timestamps, current status, and assigned team.
Report media	Protected storage object reference, ownership/authorization context, and upload metadata.
Locations	Supported campus building/location identifiers and display names.
Categories	Administrator-managed report categories and active/inactive state.
Routing rules	Category/location conditions, destination team, active state, and configuration history where required.
AI analysis	Suggested category, summary/details, suggested priority, uncertainty/confidence information, processing status, and provider metadata.
Duplicate candidates	Potentially related report identifiers, similarity signals, review state, and staff disposition.
Status history	Prior and new status, actor, timestamp, and optional staff resolution/update note.
Notifications	Notification event, recipient, channel, delivery state, and failure metadata.

5.7 Security and Service Boundaries

- All protected API operations must be authenticated and authorized on the backend.
- Student report queries must be scoped to the authenticated student.
- Staff report queries must be restricted to the teams or office queues assigned to the authenticated staff account.
- Administrator-only configuration and analytics endpoints must require administrator authorization.
- Uploaded photographs must use private storage and access mechanisms consistent with report permissions.
- AI provider credentials and other secrets must remain server-side and must never be embedded in the mobile application or committed to the repository.
- AI output must be validated before persistence or downstream use.
- The report record and backend status are authoritative; notification delivery is a separate side effect.

6. UI/UX Design

6.1 Student Experience

The student experience is centered on a prominent Snap & Report action. The design should minimize typing and avoid asking students to identify the responsible office. A photograph is the primary input, location provides operational context, and the note remains optional.

Student flow Student Home -> Snap & Report -> Capture / Select Photo -> Confirm Location -> Optional Note -> Submit -> Report Received -> My Reports / Report Details.

- Keep the Snap & Report action immediately visible from the student home screen.
- Make photo capture or selection straightforward and show a preview before submission.
- Use a simple campus building/location picker. Do not request continuous device location.
- Clearly mark the description/note as optional.
- Provide visible submission progress, an honest success/failure result, and a report identifier after successful submission.
- Show AI-generated labels as suggestions and avoid presenting model output as confirmed facts.
- Make report status and recent staff updates easy to find.

6.2 Campus Staff Experience

Staff flow Staff Dashboard -> Authorized Team Queue -> Report Details -> Review Photo / Location / AI Suggestions -> Assign or Update Status -> Resolve.

- Display only reports within the staff member's authorized team/office scope.
- Make the submitted image, location, note, category, current status, and report history easy to review.
- Visually distinguish AI suggestions from staff-confirmed information and staff actions.
- Make suggested priority and potential duplicate signals visible without forcing staff to accept them.
- Provide permitted assignment, status-update, and resolution controls.
- Record changes so report history remains understandable.

6.3 Administrator Experience

Administrator flow Administrator Dashboard -> Category Configuration / Routing Configuration / Report Trends.

- Restrict configuration functions to authorized administrators.
- Allow category and routing changes to be managed through configuration rather than requiring mobile-interface changes.
- Show aggregate trends using persisted report information and supported filters.
- Make the configured destination/team clear and avoid implying that unverified routing mappings are official CBU policy.

6.4 Report Status Model

Status	Meaning	Typical Transition Owner
Submitted	The report has been received and stored.	System / Backend
Assigned	The report has been directed to or accepted by an authorized team/person.	Routing workflow / Staff
In Progress	Authorized staff are actively addressing the issue.	Staff
Resolved	Staff have marked the reported issue as resolved.	Staff
Reopened	The student indicates that the issue remains unresolved and requests another review.	Student action under permitted rules
Closed	The reporting workflow has been completed.	Authorized staff / configured workflow

The backend must enforce permitted transitions. The project must decide whether Closed is a distinct staff action after Resolved or a later administrative state; that detail remains an implementation decision. The student-facing view must always display the current persisted state.

6.5 Accessibility and Interaction Principles

- Use clear action labels and plain language.

- Maintain sufficient contrast and readable text sizes.
- Do not use color as the only indicator of priority or status.
- Provide accessible labels for image capture, location selection, submission, and status controls.
- Show validation errors next to the relevant action and preserve user-entered information when safe to do so.
- Use explicit loading, success, empty, and failure states for network and AI-dependent operations.

7. Design Decisions, Assumptions, and Risks

7.1 Key Design Decisions

DD-01 - Photo-First Reporting. A photograph is the primary input to make campus issue reporting quick and concrete.

DD-02 - Minimum Student Input. The report flow requires only information necessary to create useful campus context. The note is optional, and the student is not required to select the responsible team.

DD-03 - AI-Assisted Classification. AI analyzes the submitted photograph and optional note to recommend a supported category and extract useful details. AI output remains advisory.

DD-04 - Smart Routing. The backend applies configurable routing logic after report information is available so that submissions can reach an appropriate operational queue.

DD-05 - Team-Based Routing. Destinations are configured teams or role-based queues, not permanently assigned individual employees. Actual CBU mappings must be validated.

DD-06 - Duplicate-Aware Reporting. Potentially similar reports may be flagged for review. Similarity alone does not automatically merge or close reports.

DD-07 - Human-Owned Operational Decisions. Authorized staff remain responsible for assigning work, responding to safety concerns, changing report status, and resolving issues.

DD-08 - Failure-Tolerant Report Submission. A valid report should be stored even when external AI analysis is unavailable. Analysis may be retried or handled through a manual-review path.

DD-09 - Emergency Boundary. LancerSnap is not an emergency-dispatch channel. The interface must direct users to appropriate emergency services or campus channels for immediate assistance.

DD-10 - Campus Reporting MVP Boundary. The MVP is limited to photo-first reporting and the supporting classification, routing, staff-processing, tracking, and analytics capabilities defined in this SDD.

7.2 Planning Assumptions and Open Decisions

ID	Open Decision	Required Clarification
OQ-01	CBU team and category routing	Confirm the operational owner for each supported report category and the team queue that should receive it.
OQ-02	Staff workflow integration	Determine whether staff will use the LancerSnap dashboard independently or whether a limited handoff to existing CBU workflows is required.
OQ-03	AI provider and model	Select the multimodal AI provider based on accuracy, privacy, cost, response time, and structured-output support.
OQ-04	Duplicate detection approach	Evaluate initial matching using category, location, time, and text; assess image similarity only if needed and feasible.
OQ-05	Campus location reference	Confirm the supported building/location list and who maintains it.
OQ-06	Notification provider	Select the notification channel/provider and define delivery-failure behavior.
OQ-07	Production single sign-on	Determine whether CBU SSO is available and approved for production use; the prototype may use the agreed Supabase authentication setup.

7.3 Implementation Risks

ID	Risk	Potential Impact	Mitigation
RISK-01	Incorrect AI Classification	AI may suggest an incorrect category.	Validate structured output, present recommendations clearly, and allow staff correction.
RISK-02	Incorrect Routing	A report may be directed to the wrong team.	Use configurable routing rules and allow authorized review/reassignment.
RISK-03	Duplicate Detection Failure	Related reports may not be identified, or unrelated reports may appear similar.	Treat duplicate matching as advisory and provide human review.
RISK-04	Incorrect Priority	A report may receive an unsuitable priority suggestion.	Keep priority advisory and apply conservative handling to potential safety hazards.
RISK-05	AI Service Failure	External AI processing may be unavailable or delayed.	Allow valid report submission and provide retry or manual-review fallback.
RISK-06	Unauthorized Data Access	Private reports or photographs may be exposed.	Enforce backend role/record authorization and private media storage.
RISK-07	Notification Failure	A student may not receive a status notification.	Persist the status independently and make it visible in the application.
RISK-08	Staff Adoption Challenges	Staff may continue to rely on existing communication methods.	Minimize additional administrative effort and align queues with existing operational teams.
RISK-09	Unconfirmed CBU Routing	The selected destination may not match the actual CBU process.	Keep mappings configurable and validate them with CBU before production.
RISK-10	Scope Expansion	Additional workflows may make the MVP unnecessarily complex.	Use the scope boundary in Section 2.4 and require human approval for scope changes.

8. References and Supporting Artifacts

The following resources support the selected implementation technologies and the project design:

- California Baptist University: https://calbaptist.edu/
- React Native documentation: https://reactnative.dev/docs/getting-started
- Expo documentation: https://docs.expo.dev/
- FastAPI documentation: https://fastapi.tiangolo.com/
- Supabase documentation: https://supabase.com/docs
- PostgreSQL documentation: https://www.postgresql.org/docs/

Supporting project artifact: LancerSnap system architecture and reporting workflow diagram (Figure 1). The approved Figma Make student and staff screens should be used as the detailed visual reference during implementation.

Appendix A. Requirement Identifier Index

Identifier Group	Range	Purpose
User Stories	US-01 to US-13	Student reporting and tracking; staff report processing; administrator routing, trends, and category management.
Requirements / Constraints	RC-01 to RC-12	Access control, AI uncertainty, human oversight, routing, duplicate handling, safety, emergency boundary, secure media, error handling, accessibility, and status transparency.
Design Decisions	DD-01 to DD-10	Photo-first/minimal-input design, AI advisory behavior, routing, duplicate handling, human decisions, failure tolerance, emergency boundary, and MVP scope.

Traceability rule: Implementation work should cite the relevant SDD identifiers and the corresponding phase/issue in the Implementation Plan. When a requirement changes, update the SDD and review affected Implementation Plan references and GitHub Issues.